STA380 Group Project Proposal

Permutation Test on World Population Development Groups (2023)

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This proposal outlines our project, including the project topic, dataset, statistical method and planned output of our shiny application.

I used an Undergraduate Project Capstone Proposal Template to write this example

1 Project Topic

Permutation Test between Developmental Groups from the World Population Prospects (2024)

Our group proposes to perform a permutation test to determine if there is a statistically significant difference in chosen demographic indicators (such as Life Expectancy, Birth Rate, or Mortality Rate) between different developmental groups (e.g., Developing vs. Developed regions[1]).

2 Simulation vs. Dataset

This project will use a **dataset** sourced from the United Nations [United Nations, Department of Economic and Social Affairs, Population Division, 2024].

- **Source:** United Nations, Department of Economic and Social Affairs, Population Division.
- **Dataset Name:** World Population Prospects 2024.
- **Link:** https://population.un.org/wpp/

We will use the "Estimates" worksheet from the WPP 2024 dataset, which contains reconstructed historical data from past censuses and surveys, ensuring our permutation test is performed on facts rather than projections of the future (such as the "Medium Variant", "Low Variant", and "High Variant" sheets).

To capture the current state of global development, we will filter the dataset for the year 2023, which represents the most recent year of estimated historical data.

The processed `Clean_data.csv` file containing the relevant indicators and developmental groups has been provided within the Proposal sub folder.

[1] Countries were classified into 'Developed' and 'Developing' groups according to the United Nations Statistics Division M49 Standard. 'Developed' includes Europe, Northern America, Australia, New Zealand, and Japan; all other regions are classified as 'Developing'.

3 Project Details

We will be using permutation test to examine whether there are statistically significant differences in selected demographic indicators between two groups:

- Group1: More developed regions
- Group2: Less developed regions

we focus on the indicator:

- Birth rate
- Life expectancy
- Mortality rate

The following outputs will be provided:

- A histogram showing the density of the birth rate in different regions
- Visualization showing the observed test statistic and the corresponding p-value
- Summary statistic for each group (mean, sample size and standard deviation)

4 User Inputs (Shiny Components)

Users will be able to modify the following:

1. Select the demographic indicator
2. Select the development groups to compare
3. Choose the year of analysis
4. Set the random seed to reproduce the test
5. Choose the number of permutations (B) used to generate the permutation distribution
6. Set the significance level (α) used to determine rejection of the null hypothesis
7. Download the permutation test results as a .csv file, including the observed test statistic, p-value, and the permutation distribution

5 Generative AI Usage Declaration

Generative AI was used in this proposal to convert the provided LaTex proposal example to a R Markdown format.

References

S. Basten. Re-examining the fertility assumptions for pacific asia in the un's 2010 world population prospects. *University of Oxford Department of Social Policy and Intervention, Barnett Papers in Social Research*, 1, 2013.

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J. N. O'Sullivan. Demographic delusions: World population growth is exceeding most projections and jeopardising scenarios for sustainable futures. *World*, 4(3):545–568, 2023. ISSN 2673-4060. doi: 10.3390/world4030034. URL https://www.mdpi.com/2673-4060/4/3/34.

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United Nations Statistics Division. Standard country or area codes for statistical use (m49), 2024. URL https://unstats.un.org/unsd/methodology/m49/. See section 1 (Project Topic) for Note on developed and developing regions.